

Exploring a Quantum-Hebbian Approach Towards Learning and Cognition

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ABSTRACT

Cognition, in all its diverse forms, can be attributed to the formation of specialized neural circuits in the brain. While in response to conscious experiences, arising due to sensory stimulation, neural assemblies are formed, conscious experience is itself an outcome of sustained synchronous firing of a neural assembly. This leads to a paradoxical situation or Neural Development Paradox - what comes first: the conscious experience or the neural assembly? To have zero-lag synchronous activity amongst neurons of an assembly, which is needed for binding individual responses into a conscious percept, we need a very rapid source of communication between them. Axonal conduction delays amounting to several tens of milliseconds cannot account for the zero-lag synchrony across spatially separated brain areas, which is required for STDP Hebbian learning to hardwire a neural assembly. Hence, an alternate means of instantaneous communication needs to be explored. According to the Penrose-Hameroff Quantum Approach towards consciousness, the microtubule cytoskeleton acts as a quantum computer whose collapsed states are conscious experiences manifesting in the form of an activated neural assembly. Neurons, whose microtubules are entangled, get connected by means of gap junctions, behaving like one giant neuron, firing in synchrony, seemingly breaking the Neural Development Paradox. By mental force of directed attention, Quantum Zeno Effect can be instigated, as suggested by Von Neumann-Stapp, causing the activated state of the brain to stabilize till Hebbian learning strengthens the synaptic connections, giving way to a quantum explanation for the formation of neural assemblies, both intra-modality as well as inter-modality. In this paper we report our efforts towards building a simulation-mathematical model of Quantum-Hebbian Learning that attempts to interface classical Hebbian learning with quantum theoretic mechanisms of conscious experience to develop neural assemblies. Under the influence of the modulatory mental force of directed attention, the model is also able to address high level issues like mind wandering, zombie modes, volition etc. The model is an attempt to study the benefits of introducing the concept of duality in the formation of neural assemblies and understand how consciousness and attention influence the development of the brain.

Key Words: Cognition, neural assemblies, quantum entanglement, quantum zeno effect, quantum neural computation, attention

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1. Introduction

All cognitive abilities of the brain arise from the activation of specialized neural assemblies that are spread across spatially separated brain areas. The formation of these well-defined

neural assemblies, in response to sensory stimulation, is the hallmark of neural development. Conscious perception is attributed to the coherent oscillations (in the gamma frequency range) of neurons of an assembly, spread across different brain regions, responding to a particular stimulus. These coherent oscillations bind the individual neural responses into a conscious percept of the stimulus (Melloni et al., 2007). While conscious experience is attributed to the synchronous firing of these neural assemblies, the exact mechanism causing this synchronous

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activity leading to the formation of neural assemblies is still being debated upon.

Learning in the brain is based on the famous Hebb Rule (Hebb, 1949) '*Cells that fire together wire together*' and Stent's modification (Stent, 1973) '*Cells that do not fire together compete for elimination*'. This means that synchronized activity strengthens synaptic connections between neurons, whereas unsynchronized activity leads to pruning and eventual elimination of the synaptic connections. Neural development involves two major factors, i. Nature, ii. Nurture. Nature refers to the epigenetic compartmentalization of the different functional areas e.g., speech, vision, auditory etc. whereas nurture refers to the environmental factors and sensory stimulation that the organism receives. The brain constantly adapts and configures itself according to the environment. Despite the heterogeneity in the stimulation that different organisms receive, the presence of a uniform pattern of synchronized activity has been noticed to precede all developing networks (Ben-Ari, 2001). However, a clear consensus on how this synchronous activity arises has still not been reached. Many researchers attribute this synchrony to cortico-cortical re-entrant connections (König *et al.*, 1993; Löwel and Singer, 1992), and to thalamo-cortical re-entrant connections (Llinas and Pare, 1998; Lumer *et al.*, 1997; Edelman, 2005). There are two potential problems with these approaches. Firstly, they assume very specific re-entrant connections to exist between cortical neurons that are far away but become part of the same neural assembly, which is unlikely in an undeveloped brain and secondly, they assume axonal conduction delays to be negligible. While in the case of nearby or intra-modality neurons, with direct synaptic connections, the impact of axonal conduction delays could be treated as negligible, but in the case of neurons in spatially separated brain areas, where axonal conduction delays could amount to several tens of milliseconds (Swadlow and Waxman, 2012), zero-lag synchrony cannot be attributed to axonal transmissions. This leads to a paradoxical situation, that we term as the Neural Development Paradox - *In the absence of any direct axonal connections, how do neurons in spatially separated brain areas synchronize?* It seems unlikely that the brain would employ different mechanisms for synchronizing neural activity at the intra-

modality and inter-modality levels and therefore there is a need to take a fresh look at this issue and search for a non-axonal rapid means of communication. Previous attempts to look for non-axonal means of communication included models based on diffusive Hebbian learning (Markan, 1996; Markan and Bhaumik, 1999), neural gas (Martinetz and Schulten, 1991), volume learning (Swindale and Mitchison, 1999), neurotrophins (Elliott and Shadbolt, 1999) etc. The shortfall of these models is that they are only able to explain nearby neural interactions, giving no account of inter modality neural interactions.

It is evident from the synchronous activity that precedes the formation of neural connections (Ben-Ari, 2001; Uhlhaas *et al.*, 2009), that there is presence of a coordinating agent that overlooks or supervises brain wide neural activity and is faster than synaptic transmission. Whether this agent is a classical entity or a quantum entity is the mystery that needs to be unraveled. Recently, two different mechanisms for feature binding have been postulated. First, a '*Hardwired*' binding of previously seen natural objects through a hierarchy of feature detectors of increasing complexity, wired by Hebbian learning principles and second, an '*on-demand*' binding regulated by attention, which acts when more arbitrary or unfamiliar features are presented. While the former system works without attention, it is challenged by spatial competition which arises when neuronal receptive fields overlap. This happens when multiple objects are presented in close vicinity. Thus, while '*on-demand*' binding always requires attention, '*hardwired*' binding only does so when receptive field competition occurs (Vanrullen, 2009). It seems nature has bestowed upon us a unique gift in the form of the ability to adapt to any unfamiliar, unnatural situations/stimulation by means of attention. Attention causes the necessary binding that leads to Hebbian learning. Once the situations are no longer unfamiliar or unnatural, because of hardwiring of neural assemblies for that stimulus, or as some call it '*accumulation of knowledge*', we tackle them rather effortlessly, even in the absence of or under distracted attention. This gives humans a huge survival advantage making us superior to all other life forms.



While attention, as a modulator for learning has been studied by many, the exact genesis of attention is still unknown. Recent research in neuroscience reveals the presence of Nucleus Basalis of Meynert (NBM), a part of the basal forebrain, which along with other neuromodulators is thought to ‘energize’ the neocortex. It has been observed that artificial activation of the NBM increases the reliability of neural responses and also leads to decreased redundancy of information processing, resembling brain states that are observed when attention is at play (Chen *et al.*, 2012; Thiele, 2009). These findings give a direct demonstration of how NBM dynamically regulates the coding of sensory inputs in the cortex. Because NBM receives inputs from both subcortical regions and prefrontal cortex, it seems to act as a relay station for both bottom-up and top-down signals to regulate sensory coding in a behaviorally relevant manner (Goard and Dan, 2009). On the basis of these experimental evidences we could postulate NBM to be a part of the central core from where attention arises and spreads to different parts of the brain. While it is agreed upon that attention is closely associated with synchronized neural activity over a large area of the brain, different schools of thought exist regarding the origin of this synchronous activity. While some attribute this synchronous activity to a classical phenomenon arising due to re-entrant connections as discussed above, interestingly, Hameroff in his quantum theory of consciousness suggests that this synchronous activity is due to quantum mechanical entanglement of microtubules. He proposes that quantum tunneling through gap junctions allows intracellular quantum states to spread among neurons whose microtubules are entangled. The depolarization of these neurons, connected by gap junctions, is perfectly synchronous and hence they fire in complete synchrony. Gap junction openings and closings evolve dynamical dendritic web topologies moving throughout axonal-dendritic networks accompanied by gamma synchrony and giving rise to consciousness (Woelf and Hameroff, 2001; Hameroff, 2007) much like the classical search light hypothesis suggested by Crick (1984) and the dynamic core as suggested by Edelman and Tononi (2000). Recent research has also brought out that gap junction or electrical coupling between sister neurons in the developing

cortex, link them into circuits that process related sensory information (Mrsic-Flogel and Bonhoeffer, 2012). Gap junctions have also been found to precede chemical synapse formation (Todd *et al.*, 2010). In the light of all these findings, gap junction mediated microtubule entanglement, suggested by Hameroff, appears to be a strong candidate for creating synchrony (Hameroff and Penrose, 2003; Hameroff, 2007). Since microtubule entanglement could happen at both nearby and far away neural interactions, this opens up the possibility to explore a quantum mechanical framework for addressing the problem of forming brain wide neural assemblies. There are many advantages of considering a quantum mechanical framework. First, the temporary connection between various neural circuits can be sufficiently fast, if we consider quantum entanglement between neurons. Secondly, these temporary connections could be easily terminated and thirdly, considering this kind of quantum interaction may help us get some insight into phenomenon like subjective experiences, free will and consciousness, which have still not been explained classically.

In this paper, in section 2, we further discuss the rationale behind exploring a quantum mechanical approach for formation of neural assemblies. In Section 3, we discuss the role of mental force of directed attention and Quantum Zeno Effect in learning and formation of neural circuits. In section 4 we describe the model of a Quantum Neural Computer, a basic building block of quantum computation in the brain. In section 5 we discuss the Quantum Hebbian Equation developed to show the quantum brain’s supervisory interaction with the classical brain and by means of MATLAB simulations we elucidate how this interaction leads to learning in the classical neural circuits. Section 6 contains the discussion.

2. Quantum Mechanics to Our Rescue?

The motivation to use quantum mechanics to demystify the brain stems from many reasons. Biological systems are made up of molecules and atoms and quantum theory is the most advanced theory to explain the interaction between them. For example, the dynamics of chemical bond formation, absorption of specific frequencies as observed in photosynthesis and single electron transfer



through DNA or proteins are all quantum effects. Recently, quantum effects have also been reported in birds' vision that help them navigate far distances using earth's magnetic field (Lambert *et al.*, 2012) and also in human smell processing or olfaction (Brookes, 2011). Therefore, there are ample experimental evidences for the presence of quantum effects in biological systems, that too at fairly high temperatures. Efforts towards demystifying how quantum information/computation processing circuits work, in the brain and in general using multidisciplinary approaches is on and is leading us towards fascinating results (Srivastava *et al.*, 2011; Srivastava *et al.*, 2013).

Penrose and Hameroff in their Orchestrated Objective Reduction (OrchOR) theory (Hameroff and Penrose, 1996, Hameroff, 2003; Hameroff, 2007) suggest that subjective consciousness is an outcome of the quantum computation in the microtubules. Recently, they have also suggested that the cytoskeleton that encases microtubules can exist in two phases, the liquid or *sol* phase and the gelation or *gel* phase (Woelf and Hameroff, 2001). These phases are related to the polymerization of actin. In the *sol* phase the cytoplasmic actin is depolymerized, thereby enabling microtubules to receive inputs from, and send output to i.e. communicate with the neuronal membrane, or interacting with the classical brain. However, in the *gel* phase the cytoskeleton acts as a shield or part of the quantum system, not allowing the microtubules to interact with the neuronal membrane. Furthermore, at physiological pH levels the terminal amino acids of tubulin extended outward into the cytoplasm, bearing negative charges, attract counter ion positive charges; this double ion layer forms a plasma phase that can screen microtubule quantum states from de-coherence (Sackett, 1995). Based on these findings the de-coherence times for microtubule bundles in actin gel have been calculated in the range of hundreds of milliseconds, that is compatible with neurophysiological processes (Hagan *et al.*, 2002). Many groups are working towards experimentally demonstrating the feasibility of other quantum phenomenon in microtubules and good results are already coming out (Bandopadhyay, 2011).

Other leading researchers like Henry Stapp (Stapp, 2001; Stapp *et al.*, 2005), also attribute higher cognition to quantum events. In his theory Stapp defines three processes (Process 1, 2 and 3) that define the causal interaction between mind and matter (brain). Process 1, the choice or question posed by the observer. Process 2, the evolution of all possible states according to the Schrödinger's equation and Process 3, the nature's response to the question posed by Process 1. He attributes the quantum events to uncertainties at the molecular level and identifies preconscious-to-conscious transitions with the collapse of a quantum wave function by Process 3, generating a neurological pattern of activity or in our words an activated neural assembly (Stapp, 2001).

In the light of these developments it seems reasonable to treat the brain as an amalgamated quantum and classical system and study further the interaction between the quantum and the classical brain to gain insight into how the classical brain might develop or evolve under the supervision of the quantum brain.

3. Quantum Ingredients to form Neural Assemblies

The formation of a neural assembly, for a particular stimulus or an abstract thought, requires three events to happen in a chronological order. First, a particular neural assembly for that sensation or thought should get selected, out of all possible neural assemblies in the brain. Second, there needs to be sustained synchronous activity of the neurons of the selected neural assembly, and third, there needs to be some change in the synaptic weights connecting the neurons of that assembly so that they become hardwired. All this happens in the presence of attention. The precise role and origin of attention have been an area of interest both in neuroscience and psychology for a long time. In the late 19th century, William James described the 'effort of attention' as a means to keep affirming and adapting the thought which, if left to itself, would slip away (James, 1890). James' description of the effort of attention on the mind-brain process is remarkably in line with what had been proposed independently, from purely theoretical considerations of the quantum physics of this process (Stapp, 2006). Interestingly, all these events can be



explained in quantum mechanical parlance by considering Quantum Wave Function Collapse, Quantum Entanglement and Quantum Zeno Effect. When the quantum wave function collapses or quantum decoherence occurs, say after a critical threshold has been reached (Penrose, 1987; Diosi, 1989), a neurological pattern of activity or 'template for action' is generated (Stapp *et al.*, 2005). According to Penrose-Hameroff's quantum approach also, collapsed states of the microtubule quantum computer are conscious experiences manifesting in the form of an activated neural assembly. Quantum Entanglement among the microtubules of neurons is responsible for the selection of a group of neurons forming a neural assembly. The neurons whose microtubules are entangled get connected by gap junctions to form one large hyper-neuron (Wolf and Hameroff, 2001). Quantum Zeno Effect, which was first demonstrated by Sudarshan and Mishra (1977), states that if a system is repeatedly measured, one can 'freeze' the evolution of the system into another state, or in other words, it can repeatedly be made to collapse to the same state. Hence, we can maintain the state of the system as long as we perform measurement on it. Quantum Zeno Effect, arising due to the mental effort of directed attention, sustains the synchronous activity of an assembly of distributed neurons connected by gap junctions, by making subsequent collapses happen to the same state of the system (Schwartz *et al.*, 2005; Georgiev, 2012; Stapp, 2012) and makes way for Hebbian learning to strengthen the synaptic connections. Direct clinical evidence of the use of the phenomenon of attention and quantum zeno effect in treating patients suffering from Obsessive Compulsive Disorder (OCD) can be found in (Schwartz and Begley, 2009). Attention can thus be said to be closely associated with the quantum brain, regulating the state to which the quantum collapses happen. When the attention is focused over a stimulus or a thought, the neural activity representing it is sustained. If attention gets diverted, the thought wanders away and the stimulus no longer elicits the same loud neural response. In the absence of attention, for a familiar stimulus, the brain invokes the previously learnt neural response, as 'hardwired binding' discussed by Vanrullen (2009) or 'Zombie Modes' discussed by Koch *et al.* (Koch and Crick, 2001). Attention also

allows us to make volitional choices, irrespective of the hardwired or learnt response. Therefore, it is evident that attention plays a major role in shaping our neural circuits. What causes us to attend to some things while ignoring others could be due to environmental factors or social reinforcement e.g., a thought pertaining to music might be important for a person hailing from a musical background, but not so important for others, likewise a child who otherwise enjoys playing games, at the time of exams, might only attend to studies and so on. But this is an area of extensive research in itself and is beyond the scope of this paper.

In the next 2 sections of this paper, we build a model based on the duality of quantum and classical interaction in the brain to understand if the above mentioned states of the brain can indeed be explained on the basis of this dual interaction. The quantum is taken to be a superior layer that regulates learning in the classical layer and the quantum layer itself is regulated by attention.

4. Quantum Neural Computation

In order to understand how the quantum computation in the brain might be working there is a need to develop mathematical and simulation models for the same. As a preliminary effort in this direction, we build a basic unit for quantum computation in the brain. We call this a Quantum Neural Computer (QNC). The term quantum neural computation was first introduced by Kak (1995). Since then many efforts have been made to combine the fields of Quantum Computing and Artificial Neural Networks, essentially to enhance the computational power of ANNs (Ezhov and Ventura, 2000; Gupta and Zia, 2001; Narayanan and Menneer, 2002). Other efforts have been made to model microtubules as quantum Hopfield networks, with an aim to test the feasibility of quantum information processing in them (Behrman *et al.*, 2006; Srivastava *et al.*, 2013). Our approach is different because we combine the two fields to further our understanding of the brain function. In our model, the function of the QNC is analogous to the microtubule cytoskeleton quantum computer. The Quantum Neural Computer takes N inputs that are either $|0\rangle$ or $|1\rangle$ and transforms them to an output state $u = \alpha|0\rangle + \beta|1\rangle$ where α and β are derived from the weights and α^2 and β^2 are



the probabilities that the output u will be a $|0\rangle$ or a $|1\rangle$. The Unitary Transformation is implemented by a weight matrix described in (Miranker, 2002) given as

$$W_i = \sum_{x=0}^{2^N-1} c_x^i \cdot x \text{ where } c_x^i \in \{0, 1\} \text{ and } x \in I.$$

For the sake of normalization, only one c_x^i is 1 for a particular input, the rest are 0. A sample weight matrix for a three input Quantum Neural Computer is shown in figure 1. The α and β are computed as below.

$$\alpha = \sum_i \delta_{a_i,0} W_i, \beta = \sum_i \delta_{a_i,1} W_i \text{ and } \alpha^2 + \beta^2 = 1.$$

While we show only one sample weight here, there are $2^3 \times 2^3 \times 2^3$ such possible weight matrices. The strength of quantum computation lies in these many possibilities of transformations and this is what we harnesses in our model.

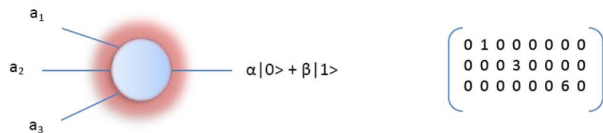
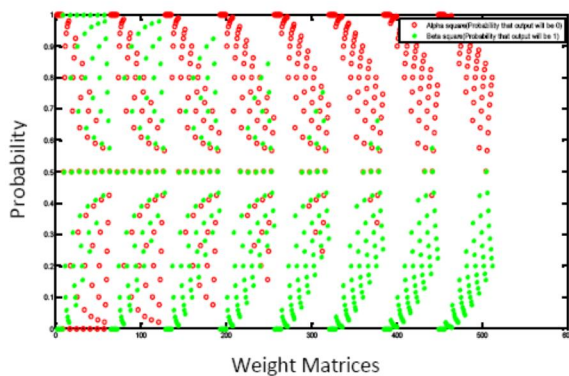


Figure 1. Left: Three input Quantum Neural Computer, Right: sample weight matrix.



The Quantum Neural Computer computes all possible output states transforming the inputs over all possible weight matrices simultaneously, by means of quantum superposition. On measurement or quantum collapse, the system reduces to a single output state that is chosen randomly from the set of all possibilities. Figure 2 (left) shows the probabilities computed over all 512 weights for the input $|010\rangle$ and Figure 2 (right) shows the way collapses are implemented. Each dot represents a collapse to a probability computed by one of the 512 weight matrices chosen at random. To draw an analogy, each weight matrix here represents a neural assembly. The weight matrix that is chosen at random represents the neural assembly that gets activated after a quantum collapse. Quantum collapses are truly random, as can be seen from Figure 2 (right). The problem that arises if conscious collapses are always random is that they fail to account for the following observations that have been made regarding experience and learning. i. Human beings always gain knowledge from their experiences, ii. They use their accumulated knowledge to take future courses of action, iii. They have a volitional choice '*free will*' to choose their actions irrespective of the accumulated knowledge.

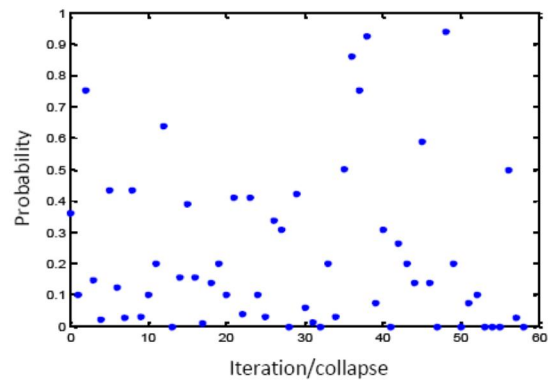


Figure 2. Left: Probabilities α and β computed for input $|010\rangle$ over all 512 weight matrices. b) Collapse implemented by means of selecting a probability (β) computed by one of the 512 weight matrices randomly.

To accommodate the above points, it becomes necessary to bring in some factor that changes with every quantum collapse, but remains stable thereafter and also influences the subsequent collapses. We call these the classical weights. To have learning in the classical brain, we need to model the impact of the quantum collapses on the classical weights. This is implemented by the Quantum Hebbian Learning Paradigm described in section 5.

5. Quantum Hebbian learning

Every quantum collapse leads to increment in knowledge which is used to decide future courses of action (Stapp, 1993; Stapp, 2001). In the human brain, the modification in synaptic weights accounts for learning or accumulation of knowledge. In order to understand how classical synaptic weights can be affected by quantum collapses we need to model their interaction. In the presence of attention ($\alpha=1$), if C is a certain classical



weight, δ the classical learning rate, and Q the impact of the current state of QNC, then δQ represents the classical accumulation of knowledge for every quantum collapse. δ is a very small value (say .001). By repeated collapses to the same state by means of attention and quantum zeno effect, the classical learning becomes sufficiently large to cause a permanent change in the synaptic weights. The equation below describes the above interaction.

$$\text{Next Collapsed State} \leftarrow \alpha Q + (1 - |\alpha Q|) * C$$

$$\text{Where, } C = C + \delta Q$$

α : attention parameter

δ : classical learning rate

Q : current state of the Quantum Neural Computer

αQ : qualia

C : classical weight (accumulated knowledge)

Here the first part of the equation αQ (named qualia) determines the quantum influence and the term $(1 - |\alpha Q|) * C$ determines the classical influence on the next collapsed state. Here α is the attention parameter that is 1 in the presence of attention and 0 in the absence of it. In the presence of attention, the first part of the equation, αQ , is large as compared to $(1 - |\alpha Q|) * C$. Therefore, the subsequent collapses of the QNC are determined by the quantum part. In a similar way, in the absence of attention, the second part of the equation, $(1 - |\alpha Q|) * C$, is large and therefore the subsequent collapses are determined by the classical part, much like the *hard-wired binding* or the *zombie* states that have been discussed previously.

Table – 1. Depicting the various states of the brain. Here, $\beta = (1 - |\alpha Q|)$.

	Non Developed Brain (C is low)	Developed Brain (C is high)
No Attention ($\alpha = 0$)	Mind Wandering	Zombie
Full Attention ($\alpha = 1$)	Quantum Zeno (when $\alpha Q > \beta C$)	Volitional Choice (when $\alpha Q > \beta C$)
	Mind Wandering (when $\alpha Q < \beta C$)	Zombie (when $\alpha Q < \beta C$)

Depending on the values of the various parameters that are involved in this equation, it leads to many possible interesting situations that are enlisted in Table I and discussed here.

Mind Wandering: In absence of attention ($\alpha = 0$) and when the classical weights have not matured or the brain has not

developed (i.e., C is very small), it leads to a state where the collapses of the QNC are always random (determined by the quantum influence). This is analogous to the state when the mind is full of random thoughts.

Zombie: In absence of attention ($\alpha = 0$), when the brain is developed (i.e., C is large), the classical accumulated knowledge determines the state to which the subsequent collapses happen e.g., when you're driving a car in an auto-pilot mode without paying attention to minor details or you respond to a familiar visual stimulus while you were attending to something else.

Quantum Zeno: In the presence of attention ($\alpha = 1$), when the classical weights have not developed i.e., $\alpha Q > \beta C$, by focused attention the subsequent collapses can be made to happen to the same state. The quantum part plays the main role. This is the state where accumulation of knowledge takes place, $C = C + \delta Q$. When C become sufficiently large, we say Hebbian learning has strengthened the synaptic connections and the neural assembly has been hardwired.

Volitional Choice (Free Will): This is a unique situation that arises where even though the classical weights are well developed i.e., C is large, the first term αQ can be larger than $(1 - |\alpha Q|) * C$, for high values of Q or Qualia. We say that these values of Q draw attention ($\alpha = 1$). The collapse of the QNC, in these cases, is determined by the quantum part rather than the classical part. This means that the brain acts in a way that overrides the accumulated knowledge, bearing analogy with the problem of volitional choice which has not been explained in neurobiological terms yet. It seems that by having a dual system (quantum + classical) in which one system has supremacy over the other, some insight into how free will could be working can be obtained. By making the classical term a difference term, $(1 - |\alpha Q|) * C$, in our model, the quantum term, αQ , always has supremacy over the classical term.

A MATLAB simulation model of the Quantum Hebbian paradigm was developed for a 3 input QNC. In order to model behavior in the different regions of the quantum Hebbian equation, an algorithm was devised. A choice register of size 10 was created. Depending on which term of the equation dominates αQ or $(1 - |\alpha Q|) * C$, the quantum part or the classical part, a certain percentage of the



choice matrix gets filled with weight matrix that dominated. A lower percentage of the choice matrix is filled with the other weight and the remaining is left random.

Subsequent choices (collapses) are made from the choice register. In this way we increase or decrease the chances of collapsing to the same state. Figure 3 shows the simulation results.

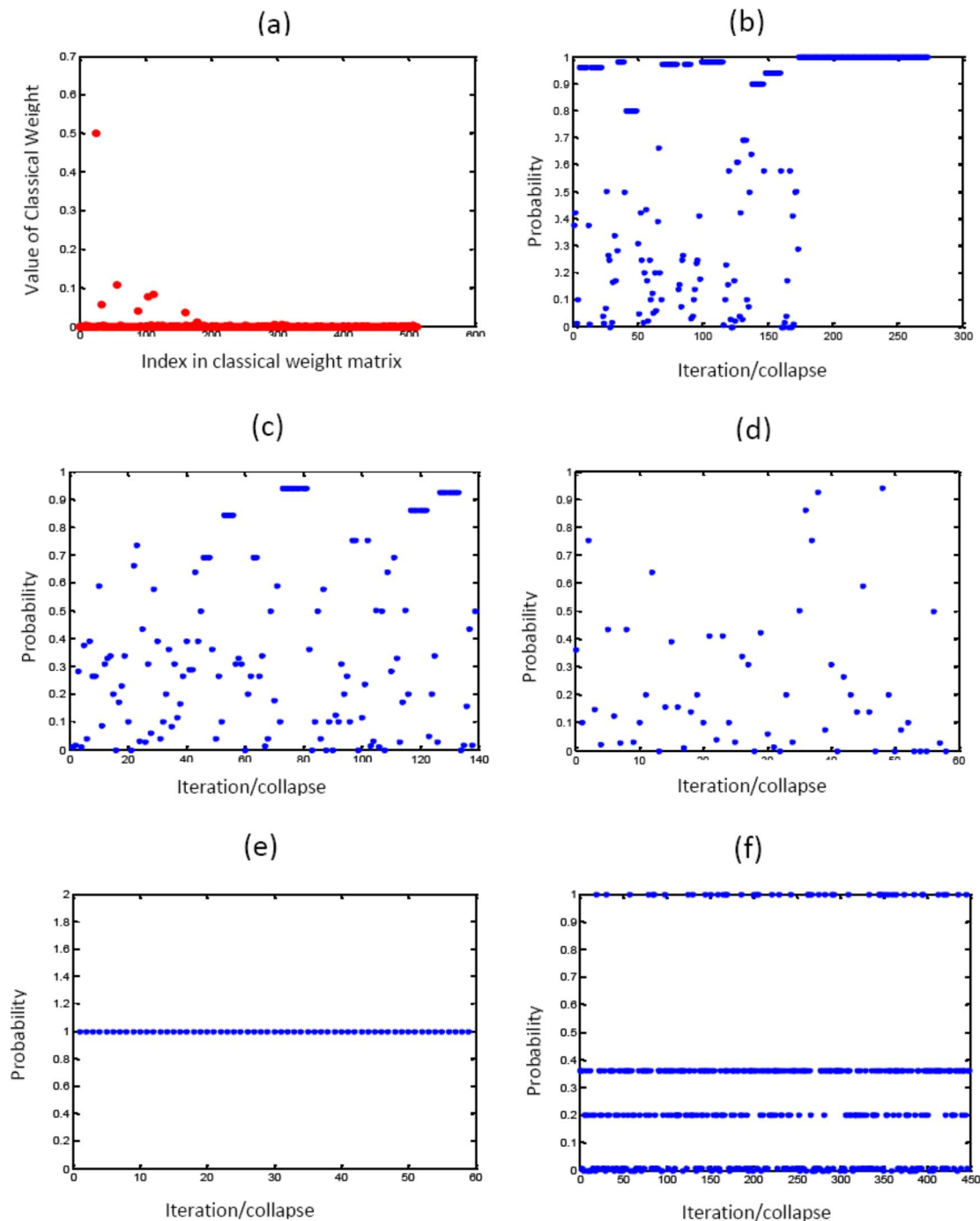


Figure 3. a). Shows the accumulation of classical weights. For a given input only one classical weight has become strong. This means out of many possibilities of neural assemblies, weights of only one neural assembly have been strengthened (*the x-axis shows the index of the classical weights and y-axis shows the value of classical weight*). b). Shows the quantum zero effect. Repeated collapses to the same state can be observed from the continuous dots. If the consecutive collapses to the same state happen for a significant period of time, we say Hebbian learning has hardwired the neural assembly. c). Represents the mind wandering state in the presence of attention. Although the collapses are mostly random, the sequences of some continuous dots show that there is some attention and quantum zero effect taking place, but it is not sufficient to hardwire the neural assembly. d). Shows the mind wandering state in the absence of attention. The plot shows that the collapses are completely random. e). Shows the zombie effect, here in the absence of attention, the collapses are determined by the classical weights that have developed the most i.e. dots corresponding to probability 1. f). Shows the free will state. Here despite of the presence of a developed classical weight (dots corresponding to probability =1), majority of the collapses are happening to a state determined by the quantum part (dots corresponding to probability .35) (*for figures b-f, x-axis represents the number of iteration and y-axis shows the probability*).



6. Discussion

A framework for Quantum and Classical interaction in the brain has been proposed. The Quantum Hebbian Equation not only indicates how quantum collapses could lead to classical learning but also determine to which state subsequent collapses will happen. With the attention parameter regulating the Quantum Neural Computer, insight into the behavior of the brain in the presence and absence of attention can be obtained. The different states of the brain like learning in presence of attention vs zombie and mind wandering and attention regulated volitional choice have been explained on the basis of this quantum and classical interaction. Recent experimental findings (Vanrullen, 2009; Koch and Crick, 2001; Schwartz and Begley, 2009) confirm the presence of these states and lend support to the results of our model. Quantum collapses at critical thresholds lead to conscious experience (Penrose, 1987; Diosi, 1989). Our model is based on the effect of these quantum (conscious) collapses on the classical brain and hence makes way to explore not only the formation of neural assembles but could also help in deciphering other aspects of information processing and cognition in the brain. This is a small but significant step towards understanding how introduction of duality, in terms of quantum and classical interaction, where quantum has a supervisory role, gives us a clearer insight into the evolution of different states of the brain. The model highlights the role of attention in developing our neural circuits and shaping our personalities and therefore is a preliminary step towards understanding how attention and consciousness lay a blueprint for the development of the brain.

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References

- Bandopadhyay A. Quantum aspects of microtubule: Direct experimental evidence for the existence of quantum states in microtubule. Towards a science of consciousness. May 2-8, 2011, Sweden.
- Behrman EC, Gaddam K, Steck JE and Skinner SR. Microtubules as a quantum Hopfield network. In *The Emerging Physics of Consciousness* (pp. 351-370). Springer, Berlin-Heidelberg, 2006.
- Ben-Ari Y. Developing networks play a similar melody. *Trends in Neuroscience* 2001; 24: 353-360.
- Brookes J. Quantum processes as a mechanism in olfaction for smell recognition? *Bulletin of the American Physical Society* 2011; 56.
- Crick F. Function of the thalamic reticular complex: the searchlight hypothesis. *Proceedings of the National Academy of Sciences* 1984; 81(14): 4586-4590.
- Chen N, Sugihara H, Sharma J, Perea G, Petravic J, Le C and Sur M. Nucleus basalis-enabled stimulus-specific plasticity in the visual cortex is mediated by astrocytes. *Proceedings of the National Academy of Sciences* 2012; 109(41): E2832-E2841.
- Diósi L. Models for universal reduction of macroscopic quantum fluctuations. *Physical Review A* 1989; 40(3): 1165.
- Edelman GM and Tononi G. Reentry and the dynamic core: neural correlates of conscious experience. *Neural correlates of consciousness: Empirical and Conceptual Questions*. pp.139-151, 2000.
- Edelman GM. *Wider than the sky: A revolutionary view of consciousness*. ePenguin, 2005.
- Elliott T and Shadbolt NR. A Neurotrophic Model of the Development of the Retinogeniculocortical Pathway Induced by Spontaneous Retinal Waves. *Journal of Neuroscience* 1999; 19(18): 7951-7970.
- Ezhov A and Ventura D. Quantum neural networks. In Ed. N. Kasabov, Editor, *Future Directions for Intelligent Systems and Information Science*. Physica-Verlag, 2009.
- Georgiev D. Mind Efforts, Quantum Zeno Effect and Environmental Decoherence. *NeuroQuantology* 2012; 10 (3): 374-388.
- Goard M and Dan Y. Basal forebrain activation enhances cortical coding of natural scenes. *Nature Neuroscience* 2009; 12(11): 1444-1449.
- Gupta S and Zia RKP. Quantum neural networks. *Journal of Computer and System Sciences* 2001; 63(3):355-383.
- Hagan S, Hameroff SR and Tuszyński. Quantum computation in brain microtubules: decoherence and biological feasibility. *Physical Review E* 2002; 65(6): 061901.
- Hameroff S. Orchestrated Reduction of Quantum Coherence in Brain Microtubules: A Model for Consciousness. *NeuroQuantology* 2007; 5(1):1-8.
- Hameroff SR and Penrose R. Conscious Events as Orchestrated Spacetime Selections. *NeuroQuantology* 2003; 1: 10-35.
- Hameroff SR and Penrose R. Conscious events as orchestrated space-time selections. *Journal of Consciousness Studies* 1996; 3(1): 36-53.
- Swadlow HA and Waxman SG. Axonal conduction delays. *Scholarpedia* 2012; 7(6):1451. doi:10.4249/scholarpedia.1451,
- Hebb DO. *The organization of behavior: A neuropsychological theory*, Wiley, 1949.
- James W. *The Principles of Psychology*, with introduction by George A. Miller, Harvard University Press, 1890.
- Kak S. On Quantum Neural Computing. *Information Sciences* 1995; 83: 143-163.
- Koch C and Crick FC. On the zombie within. *Nature* 2001; 411(6840): 893-893.



- König P, Engel AK, Löwel S and Singer W. Squint affects synchronization of oscillatory responses in cat visual cortex. *European Journal of Neuroscience* 1993; 5: 501-508.
- Lambert N, Chen YN, Cheng YC, Li CM, Chen YG and Nori F. Quantum Biology, *Nature Physics* 2012; 9: 10-18.
- Llinas R and Pare D. Coherent oscillations in specific and nonspecific thalamo-cortical networks and their role in cognition. In *Thalamus*, E. G. Jones, Ed. (Amsterdam, Elsevier), 1998: 501-516.
- Löwel S and Singer W. Selection of intrinsic horizontal connections in the visual cortex by correlated neuronal activity. *Science* 1992; 255(5041): 209-212.
- Lumer ED, Edelman, GM and Tononi G. Neural dynamics in a model of the thalamocortical system. I. Layers, loops and the emergence of fast synchronous rhythms. *Cerebral Cortex* 1997; 7(3): 207-227.
- Markan CM. Sequential development of orientation and ocular dominance maps: Reaction-diffusion approach (PhD Thesis), Department of Electrical Engineering, IIT Delhi, 1996.
- Markan CM, Bhaumik B. Diffusive Hebbian model for orientation map formation. *International Joint Conference on Neural Networks* 1999; 1: 88-91.
- Martinetz T and Schulten K. A Neural Gas Network Learns Topologies. *Elsevier Neural Networks* 1991; pp.397-402.
- Melloni L, Molina C, Pena M, Torres D, Singer W and Rodriguez E. Synchronization of neural activity across cortical areas correlates with conscious perception. *The Journal of Neuroscience* 2007; 27(11): 2858-2865.
- Miranker WL. Quantum Neurons. Tech report, Univ. of Yale, Yale/DCS/tr1234, 2002.
- Misra B and Sudarshan ECG. The Zeno's paradox in quantum theory. *Journal of Mathematical Physics* 1977; 18: 756.
- Mrsic-Flogel, TD and Bonhoeffer T. Neuroscience: Sibling neurons bond to share sensations. *Nature* 2012; 486(7401): 41-42.
- Narayanan A, Menneer T. Quantum artificial neural network architectures and Components. *Information Sciences* 2002; 124(1-4): 231-255.
- Penrose R. Newton, quantum theory and reality. In *300 years of gravity* (Ed. S. W. Hawking & W. Israel). Cambridge University Press, 1987.
- Sackett DL. Structure and function in the tubulin dimer and the role of the acidic carboxyl terminus. *Sub-cellular Biochemistry* 1995; 24: 255.
- Schwartz JM and Begley S. *The mind and the brain*. Harper Collins e-books, 2009.
- Schwartz JM, Stapp HP and Beauregard M. Quantum physics in neuroscience and psychology: a neurophysical model of mind-brain interaction. *Philosophical Transactions of the Royal Society B: Biological Sciences* 2005; 360(1458): 1309-1327.
- Srivastava DP, Sahni V and Satsangi PS. Graph-theoretic quantum system modelling for information/computation processing circuits. *International Journal of General Systems* 2011; 40(8), 777-804.
- Srivastava DP, Sahni V and Satsangi PS. Graph Theoretic Quantum System Modelling for Neuronal Microtubules as Quantum Hopfield Networks, Toward a science of consciousness. March 3-9, 2013, Dayalbagh, Agra, India.
- Stapp HP. *Mind, matter and quantum mechanics*. Berlin: Springer, 1993.
- Stapp HP. Quantum theory and the role of mind in nature. *Foundations of Physics* 2001; 31(10): 1465-1499.
- Stapp HP. Reply to a Critic: "Mind Efforts, Quantum Zeno Effect and Environmental Decoherence". *NeuroQuantology* 2012; 10 (4): 601-605.
- Stapp HP. Quantum Interactive Dualism, II: The Libet and Einstein-Podolsky-Rosen Causal Anomalies. *Erkenntnis* 2006; 65(1): 117-142.
- Stent GS. A physiological mechanism for Hebb's postulate of learning. *Proceedings of the National Academy of Sciences* 1973; 70(4): 997-1001.
- Swadlow HA and Waxman SG. Axonal conduction delays. *Scholarpedia* 2012; 7(6):1451. doi:10.4249/scholarpedia.1451.
- Swindale NV and Mitchison GJ. Can Hebbian Volume Learning Explain Discontinuities in Cortical Maps? *Neural Computation* 1999; 11: 1519-1526.
- Todd KL, Kristan WB and French KA. Gap junction expression is required for normal chemical synapse formation. *The Journal of Neuroscience* 2010; 30(45): 15277-15285.
- Thiele A. Optimizing brain processing. *Nature Neuroscience* 2009; 12(11): 1359-1360.
- Uhlhaas PJ, Roux F, Rodriguez E, Jagiela AR and Singer W. Neural synchrony and the development of cortical networks. *Trends in Cognitive Sciences* 2009; 14(2): 72-80.
- Vanrullen R. Binding hardwired versus on-demand feature conjunctions. *Visual Cognition* 2009; 17(1-2): 103-119.
- Woolf NJ and Hameroff SR. A quantum approach to visual consciousness. *Trends in Cognitive Sciences* 2001; 5(11): 472-478.

